Understanding Time In Natural Language:

-- Structured Learning, Common Sense, And Data Collection

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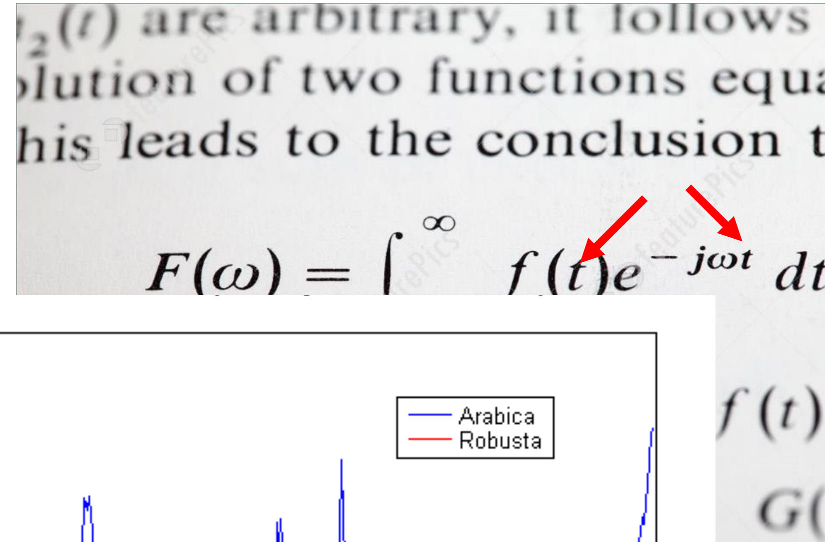
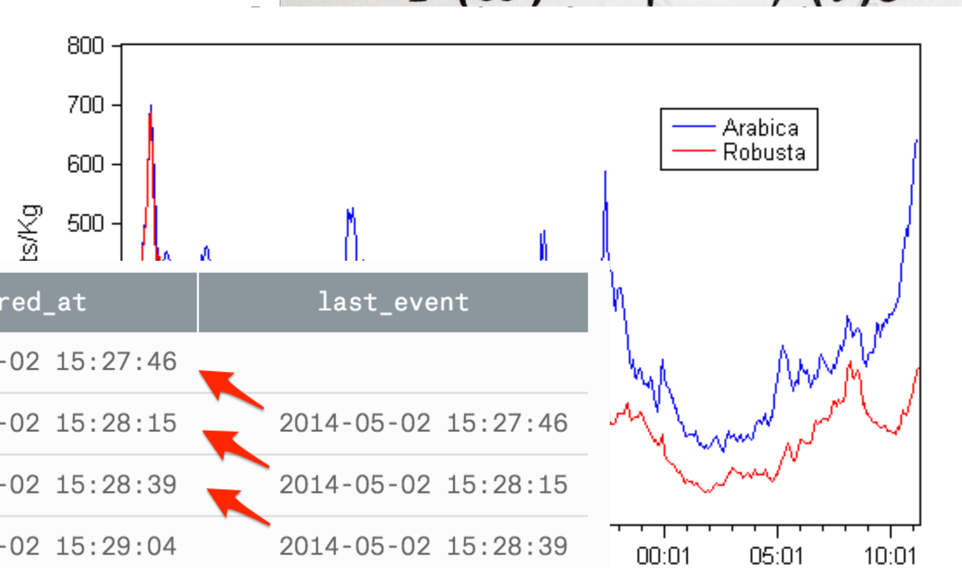
Cognitive Computation Group

Time Is A Crucial Dimension

- Signal processing
- Time series analysis
- Data mining
- …





user_id	occurred_at	last_event
8	2014-05-02 15:27:46	
8	2014-05-02 15:28:15	2014-05-02 15:27:46
8	2014-05-02 15:28:39	2014-05-02 15:28:15
8	2014-05-02 15:29:04	2014-05-02 15:28:39
8	2014-05-02 15:29:31	2014-05-02 15:29:04
8	2014-05-02 15:30:07	2014-05-02 15:29:31
8	2014-05-02 15:30:42	2014-05-02 15:30:07
8	2014-05-02 15:31:10	2014-05-02 15:30:42
8	2014-05-02 15:31:34	2014-05-02 15:31:10

Temporal Information Is Crucial In Language

People were angry → ***Police used tear gas.***

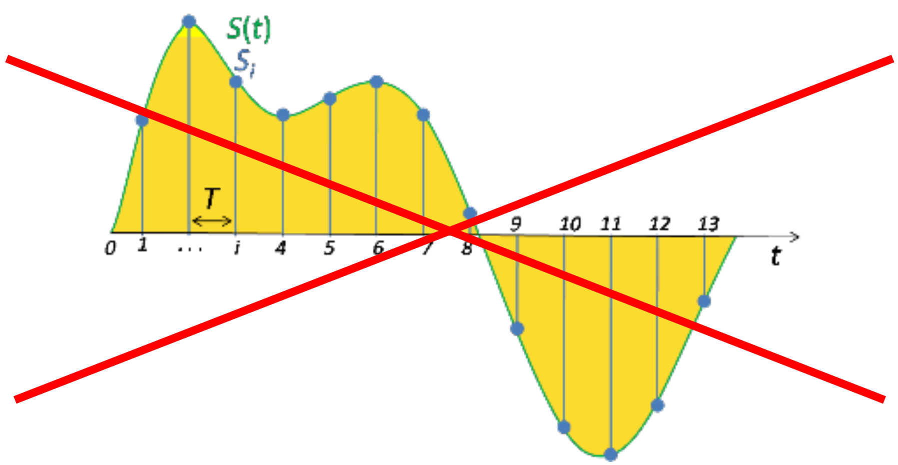
Let "A → B" represent A is temporally BEFORE B

- Story 1: People were angry → Police used tear gas
- Story 2: Police used tear gas → People were angry

Understanding Time In Natural Language

- … answers the fundamental question of when something happens/happened
- A key difference to other domains: **lack of gold timestamps**
- Temporal information is often expressed via temporal relations

"TempRel"



TEMPORAL RELATION (TEMPREL)

BEFORE

BE_INCLUDED

I met with him before leaving for Paris on Thursday.

Other temporal relations: AFTER, INCLUDES, and EQUAL

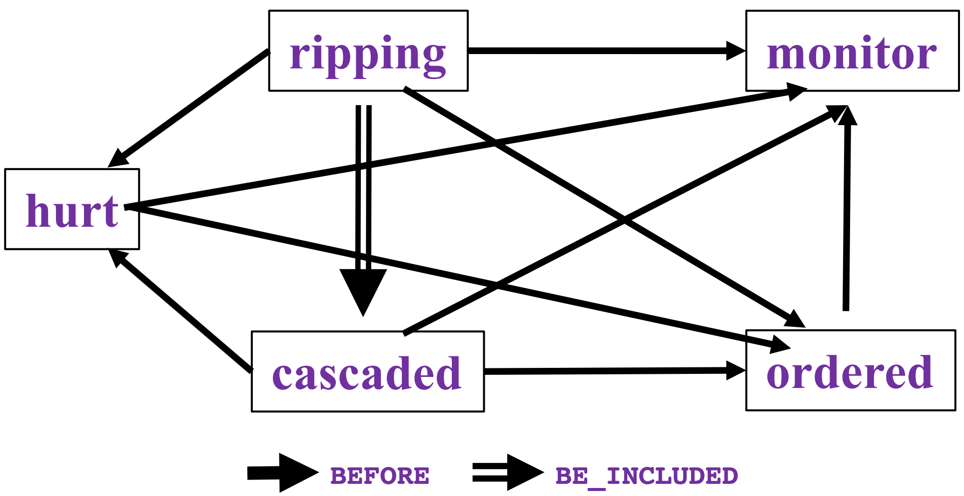
My research has mainly focused on temporal relation (TempRel) extraction (given events).

TEMPORAL RELATIONS AMONG MANY EVENTS

TEMPORAL GRAPHS

- In other words, my focus is to extract those edges in temporal graphs.

- In Los Angeles, that lesson was brought home today when tons of earth **cascaded** down a hillside, **ripping** two houses from their foundations. No one was **hurt**, but firefighters **ordered** the evacuation of nearby homes and said they'll **monitor** the shifting ground until March 23rd.



This task is difficult:

Temporal relation extraction	Literature F1 (%)
w/ gold events	low 50's
w/o gold events	low 30's

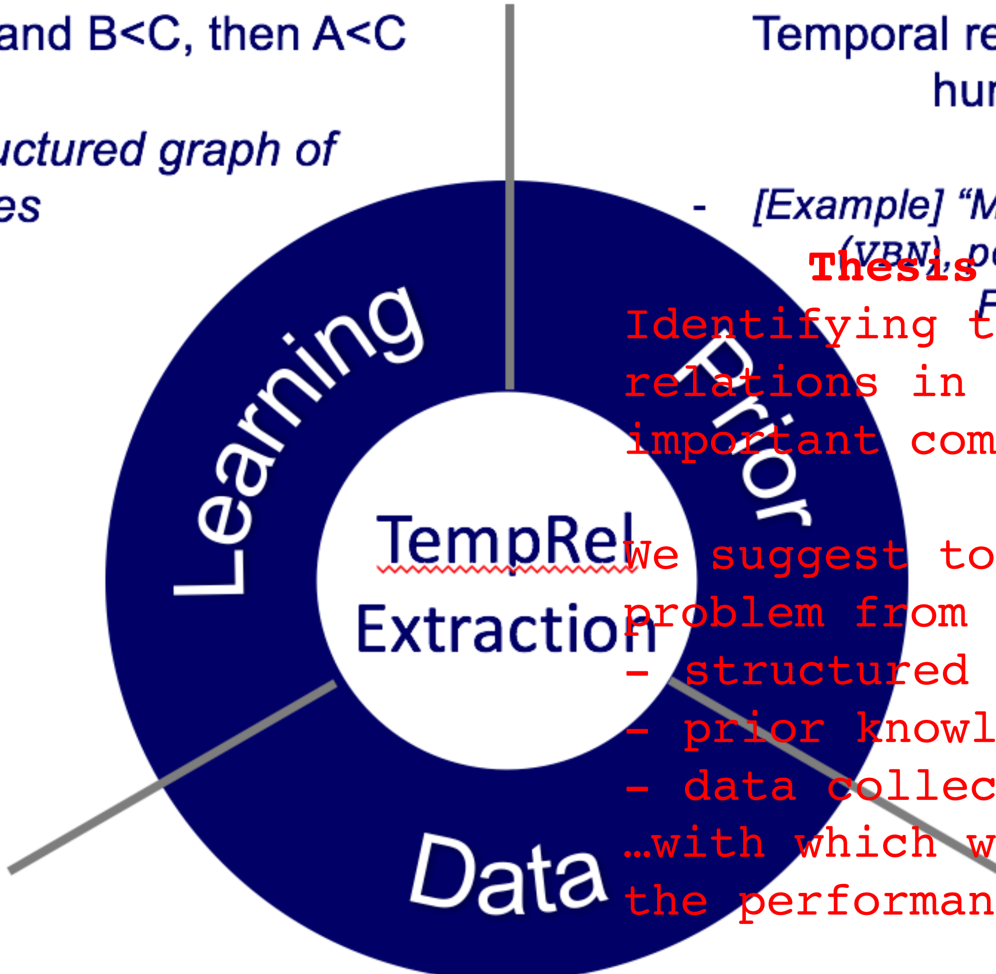
CHALLENGES OF TEMPORAL RELATION EXTRACTION

Transitivity: If A<B and B<C, then A<C

- *[Structure] a structured graph of interrelated nodes*

Temporal relations heavily rely on human's prior knowledge

- *[Example] "More than 10 people have (VBN), police said. A car (VBD) on Friday in a group of men."*



Thesis statement

Identifying temporal relations in text is an important component.

We suggest to address this problem from 3 perspectives:
- structured learning
- prior knowledge
- data collection

…with which we've improved the performance by ~20%.

Lack of high-quality data

- *[Intensive labor] Given n events, there're $O(n^2)$ edges*
- *[Difficulty] Not only before/includes, but also involves whether a relation exists*

OUTLINE

- Preliminaries: Temporal Relation and Temporal Graph
- Challenges of Temporal Relation Extraction
- A Structured Learning Approach [EMNLP’17, *SEM’18]
- Injection of Human Prior Knowledge [ACL’18a, NAACL’18]
- A New Data Annotation Scheme via Crowdsourcing [ACL’18b]
- Proposed Remaining Work

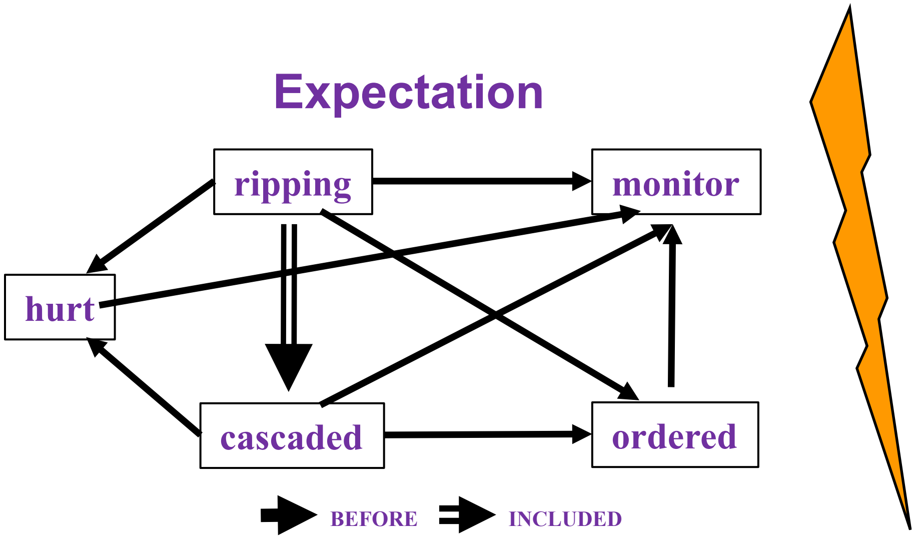
STRUCTURED LEARNING FOR TEMPREL EXTRACTION

[EMNLP'17] **Qiang Ning**, Zhili Feng, and Dan Roth. "A Structured Learning Approach to Temporal Relation Extraction.“

[*SEM'18] **Qiang Ning**, Zhongzhi Yu, Chuchu Fan, and Dan Roth. "Exploiting Partially Annotated Data in Temporal Relation Extraction."

Temporal Graphs Are Structured

- Due to transitivity, TempRels are highly <u>interrelated</u>, but existing methods learn models by considering a <u>single</u> pair at a time.



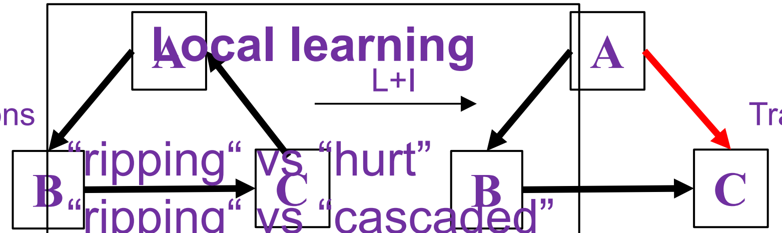
Existing methods

“ripping“ vs “hurt”
“ripping“ vs “cascaded”
“ripping“ vs “monitor”
…

PREVIOUS APPROACHES

- Local methods: Local learning + Local Inference
  - Learn models that make pairwise decisions between each pair of events
  - Transitivity is not enforced in inference

There may exist loops in the predictions



Transitivity is enforced via ILP

Local learning

“ripping“ vs “hurt”

“ripping“ vs “cascaded”

“ripping“ vs “monitor”

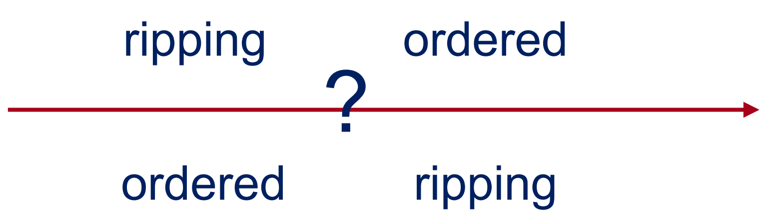
- L+I: Local learning + Global Inference
  - On top of pre-trained local models, perform a global inference over the entire graph
  - Transitivity guaranteed in inference

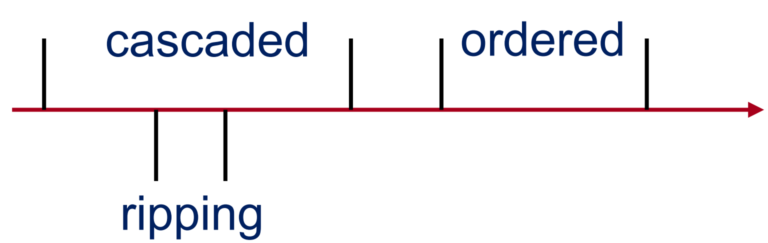
Local Learning Is Not Sufficient

- Neither local methods nor L+I methods account for the structural constraints in the learning phase.
- But information from other events is often necessary.

tons of earth cascaded down a hillside,

- …**ripping** two houses…firefighters **ordered** the evacuation of nearby homes…
  - Q: (ripping, ordered)=? *(A: by annotation, it's BEFORE)*
  - So, training on this instance using local information leads to overfitting
- (ripping, cascaded)=INCLUDED and (cascaded, ordered)=BEFORE → (ripping, ordered)=BEFORE





Global/Structured Learning

Local Learning

For each (x, y)

$$\hat{y} = sgn(w^T x)$$

If $y \neq \hat{y}$

Update w

- (x, y): feature and label for a single pair of events
- When learning from (x, y), the algorithm is unaware of decisions with respect to other pairs.

Global Learning

For each (X, Y)

$$\boxed{\hat{Y} = \underset{Y \in \mathcal{C}}{\operatorname{argmax}}\, W^T X}$$

If $Y \neq \hat{Y}$

Update W

- (X, Y): features and labels from a whole document (e.g., concatenation)
- $Y \in \mathcal{C}$: Enforce transitivity through constraint $\mathcal{C}$.

Solving The Inference Step In Global Learning

- Integer Linear Programming (ILP)

softmax: P{(i,j)=r}

boolean var: (i,j)=r

$$\hat{I} = arg \max_{I} \sum_{i<j} \sum_{r} f_r(ij) \boxed{I_r(ij)}$$

s.t. $\forall i, j, k$

$$\sum_{r} I_r(ij) = 1$$

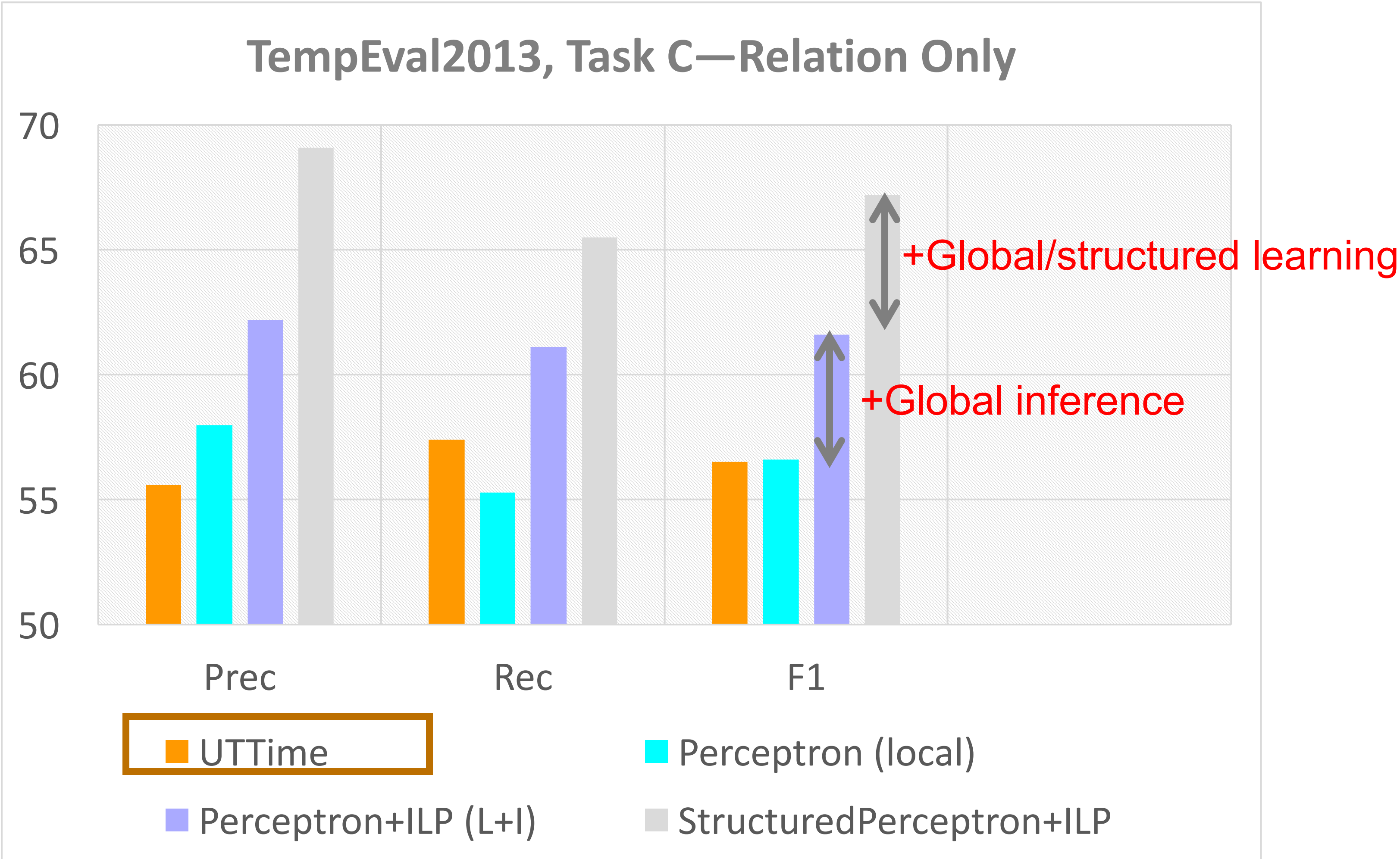
Uniqueness

$$I_{r1}(ij) + I_{r2}(jk) - I_{r3}(ik) \leq 1$$

Transitivity

- We're maximizing the score of an entire graph while enforcing those structural constraints.

RESULTS

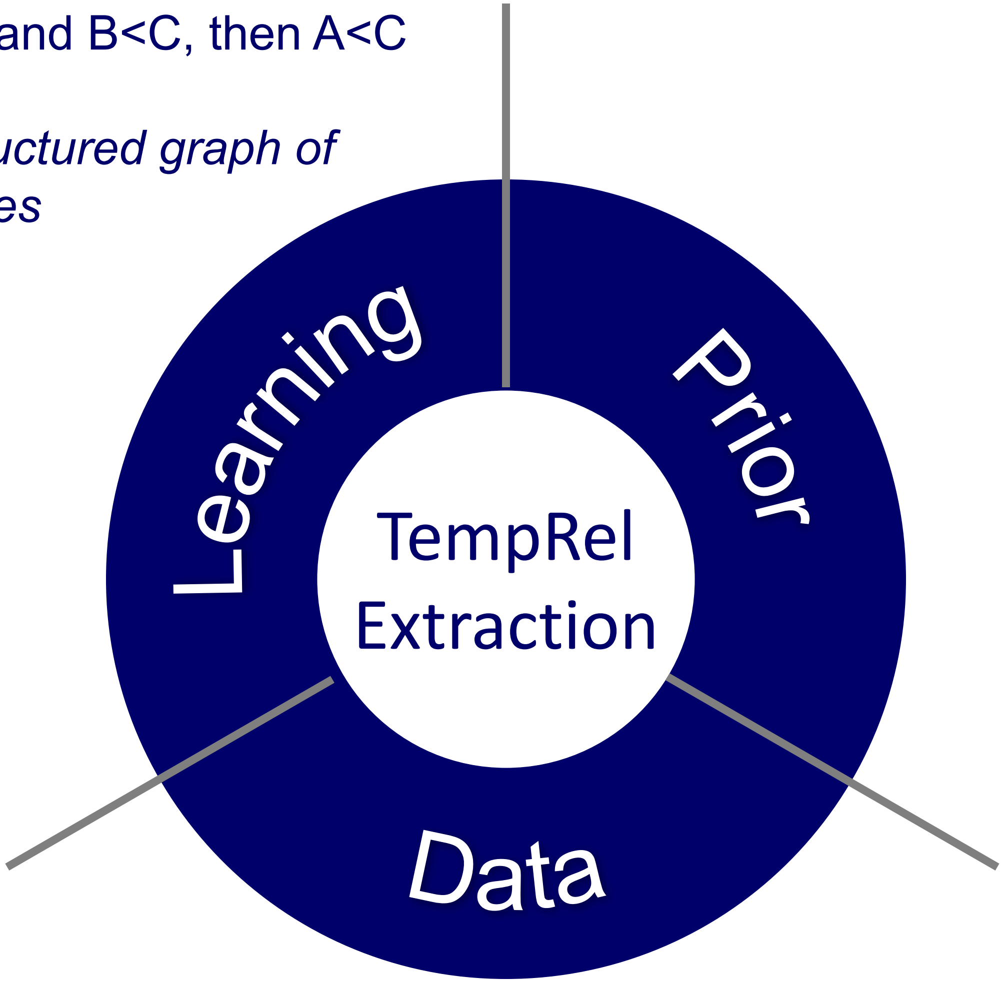


*[UTTime] Laokulrat et al., UTTime: Temporal relation classification using deep syntactic features, *SEM'13*

Challenges of Temporal Relation Extraction

Transitivity: If A<B and B<C, then A<C

- *[Structure] a structured graph of interrelated nodes*



INJECTION OF HUMAN PRIOR KNOWLEDGE

[ACL'18a] **Qiang Ning**, Zhili Feng, Hao Wu, and Dan Roth. "Joint Reasoning for Temporal and Causal Relations."

[NAACL'18] **Qiang Ning**, Hao Wu, Haoruo Peng, and Dan Roth. "Improving Temporal Relation Extraction with a Globally Acquired Statistical Resource."

When Event Content Is Missing

- More than 10 people have (**event1:**), police said. A car (**event2**:) on Friday in a group of men.
  - Part-of-speech tags: e1 = VBN, e2 = VBD
- Question: (e1,e2)=`BEFORE` or `AFTER`?
- This turns out to be difficult because we cannot use our understanding of those verbs.

When Event Content Is Present

- More than 10 people have (**event1: died**), police said. A car (**event2**: **exploded**) on Friday in a group of men.
- Explanation: Humans have an "instinct" of which event "usually" happens before another.
- Sometimes, decisions are purely based on this prior knowledge.

WHEN EVENT CONTENT IS PRESENT

- (event1: died)
  (event2: exploded)
- (event3: ask) (event4: help)

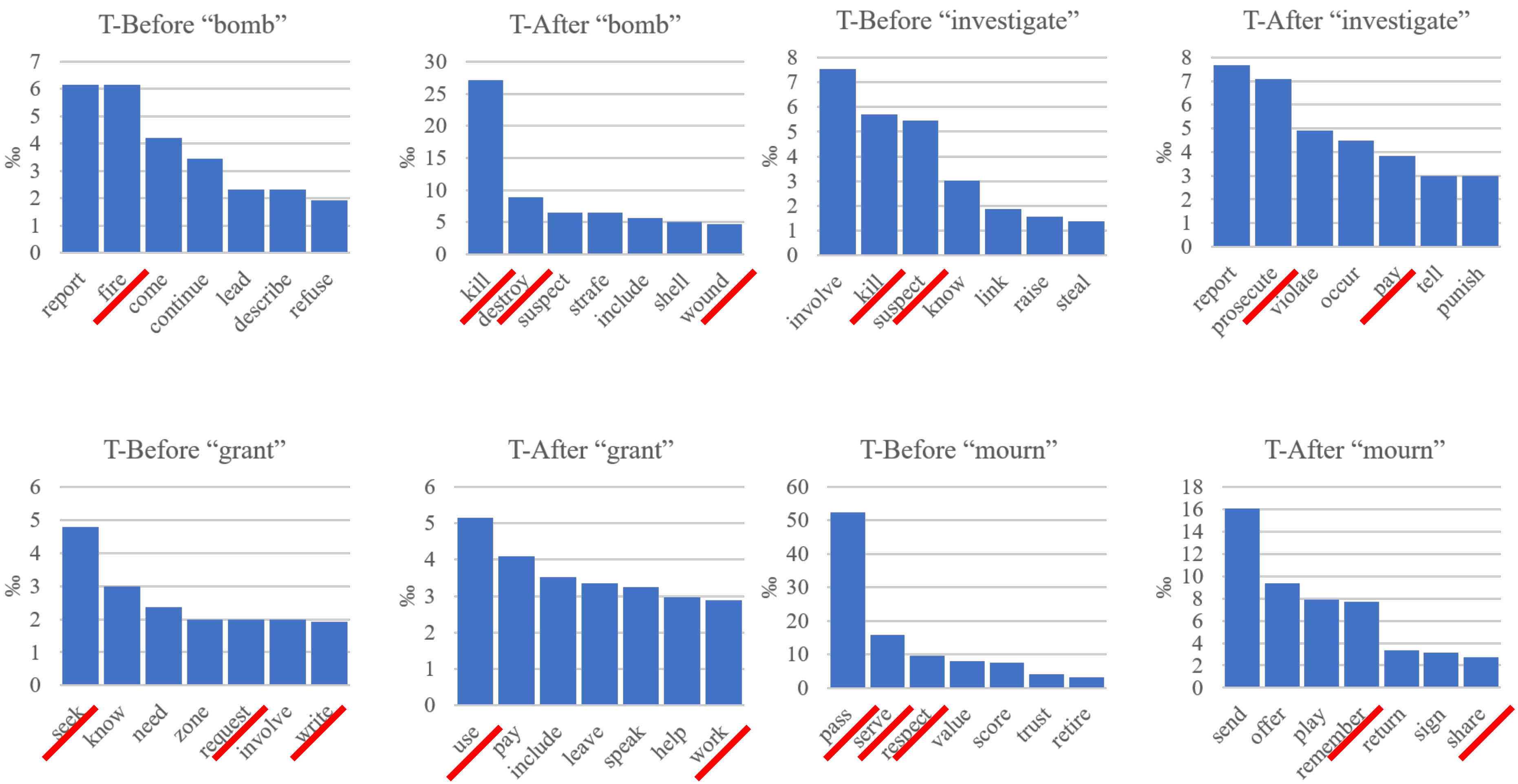
- Sometimes, decisions are purely based on this prior knowledge.
- Sometimes, pure prior is not enough, and we need to combine with the local context.
  - (Event5: dislike) (Event6: stop)
- Either way, we want to incorporate this prior in our system.

TEMPROB: TEMPORAL RELATION PROBABILISTIC KNOWLEDGE BASE

- How do we get this prior knowledge?
  - New York Times 1987-2007, #Articles~1M
  - Run a TempRel extractor on top of it
  - Simply by counting

Example pairs		Before (%)	After (%)
Accept	Determine	42	26
Ask	Help	86	9
Attend	Schedule	1	82
Accept	Propose	10	77
Die	Explode	14	83

TEMPROB: EVENT DISTRIBUTIONS



Injection Of This Prior Knowledge: Learning

- Let $C(v_i, v_j, r)$ be the number of appearances of (v_i, v_j) in TemProb that are classified to be relation r.

- For each pair of verb events,

$$h_r(v_i, v_j) = \frac{C(v_i, v_j, r)}{\sum_{r'} C(v_i, v_j, r')}$$

is the prior probability of them being relation r.

- We can add t

Example pairs		Before (%)	After (%)
Accept	Determine	42	26
Ask	Help	86	9
Attend	Schedule	1	82
Accept	Propose	10	77
Die	Explode	14	83

Injection Of This Prior Knowledge: Inference

- We can further add this prior as regularization terms.

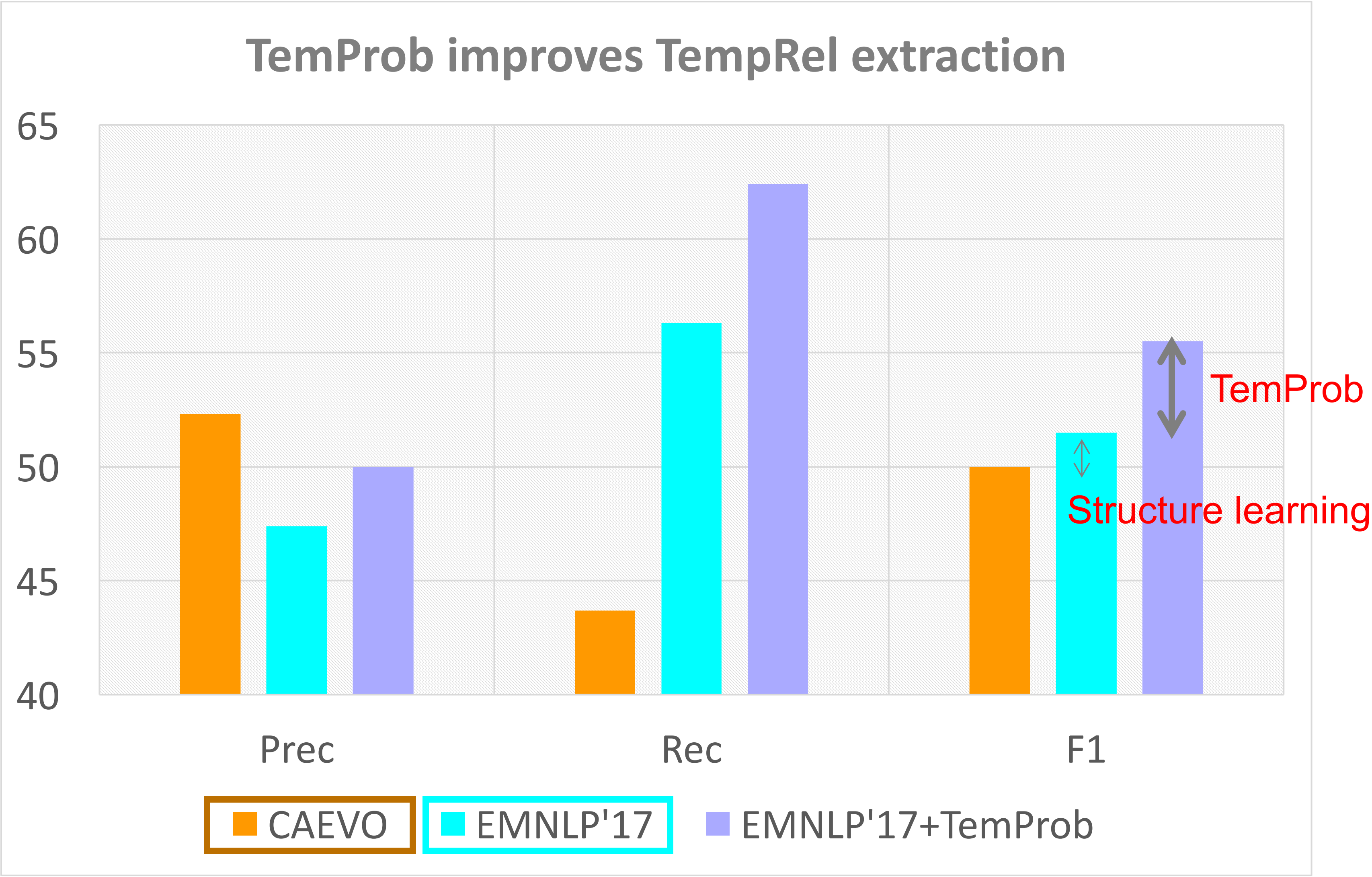
$$\hat{I} = arg\max_{I} \sum_{i<j} \sum_{r} (f_r(ij) + \boxed{h_r(ij))} I_r(ij)$$

s.t. $\forall i, j, k$

$$\sum_{r} I_r(ij) = 1$$ Uniqueness

$$I_{r1}(ij) + I_{r2}(jk) - I_{r3}(ik) \leq 1$$ Transitivity

Results



TemProb improves TempRel extraction
65
60
55
50
45
40
Prec
Rec
F1
TemProb
Structure learning
CAEVO
EMNLP'17
EMNLP'17+TemProb

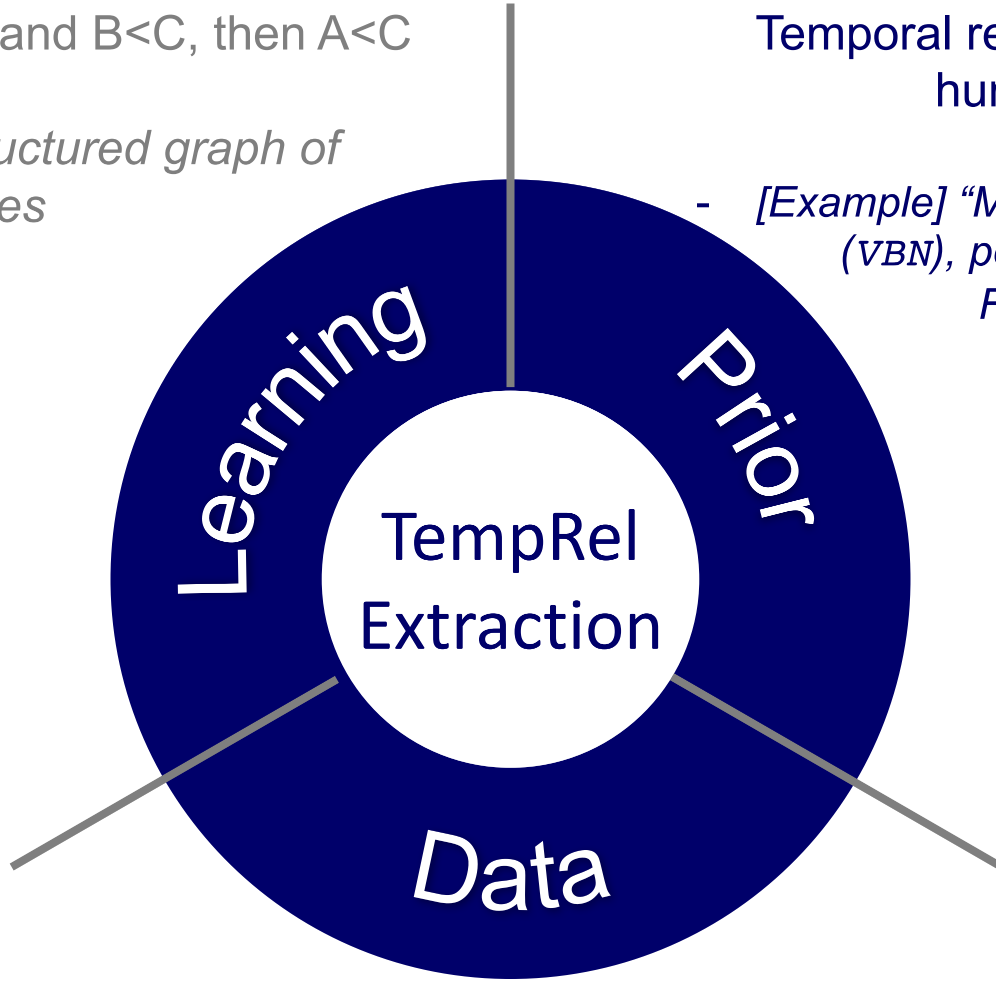
CHALLENGES OF TEMPORAL RELATION EXTRACTION

Transitivity: If A<B and B<C, then A<C

- *[Structure] a structured graph of interrelated nodes*

Temporal relations heavily rely on human's prior knowledge

- *[Example] "More than 10 people have (VBN), police said. A car (VBD) on Friday in a group of men."*



TEMPROB: INTERESTING CASES

Appears earlier in text   Appears later in text

Example Pairs		#T-Before	#T-After
chop.01	taste.01	133	8
concern.01	protect.01	110	10
conspire.01	kill.01	113	6
debate.01	vote.01	48	5
dedicate.01	promote.02	67	7
fight.01	overthrow.01	98	8
achieve.01	desire.01	7	104
admire.01	respect.01	7	121
clean.02	contaminate.01	3	82
defend.01	accuse.01	13	160
die.01	crash.01	8	223
overthrow.01	elect.01	3	100

RESULTS

- TemProb: Improving Temporal Relation Extraction
  - 2. Add the prior probabilities $\{h_r\}$ as features in learning
  - 3. Add regularization terms during inference

No.	System	P	R	F1
1	Baseline	48.1	44.4	46.2
2	**+features**	**50.6**	**52.0**	**51.3**
3	**+regularization**	**51.3**	**53.0**	**52.1**

Data Annotation via Crowdsourcing

[ACL'18b] **Qiang Ning**, Hao Wu, and Dan Roth. "A Multi-Axis Annotation Scheme for Event Temporal Relations."

We Are Almost Approaching The "Upper Bound"

- The performance of temporal relation systems has come to a bottleneck. Even if we address this problem from <u>structured learning</u> and <u>prior knowledge</u>, it's still very low.
- Performance is "upper bounded" by the quality of datasets (i.e., inter-annotator agreements)
  - TB-Dense: Cohen's Kappa 56%~64%
  - RED: F1<60%
  - EventTimeCorpus: Krippendorff's Alpha~60%
  - …
- This means that the annotation task was even difficult for human annotators.

What we achieved:

- **276 docs:** Annotated the 276 documents from TempEval3
- **1 week:** Finished in about one week (using crowdsourcing)
- **80%:** IAA improved from literature's 60% to 80%

- Re-thinking **TempRels** between events
  - … led to re-defining the task and the corresponding annotation scheme, in order to make it feasible at least to humans

Temporal Structure Modeling: Existing Annotation Schemes

- *"Police **tried** to **eliminate** the pro-independence army and **restore** order. At least 51 people were **killed** in clashes between police and citizens in the troubled region."*
- Task: to annotate the TempRels between the **bold** faced events

- Existing Scheme 1: General graph modeling *(e.g., TimeBank, ~2003)*
  - Annotators *freely* add TempRels between those events.
  - It's *inevitable* that some TempRels will be missed.
  - E.g., only one relation between "**eliminate**" and "**restore**" is annotated in TimeBank, while other relations such as "**tried**" is before "**eliminate**" and "**tried**" is also before "**killed**" are missed.

Temporal Structure Modeling: Existing Annotation Schemes

- *"Police **tried** to **eliminate** the pro-independence army and **restore** order. At least 51 people were **killed** in clashes between police and citizens in the troubled region."*

- Existing Scheme 2: Chain modeling *(e.g., TimeBank-Dense ~2014)*
  - *All* event pairs are presented, one-by-one, and an annotator *must* provide a label for each of them.
  - *No* missing relations anymore.
  - *Motivation*: In the physical world, time is one dimensional, so we should be able to temporally compare any two events.
  - *However*, some pairs of events are very *confusing*, resulting in *low* agreement.
  - E.g., what's the relation between ***restore*** and ***killed***?

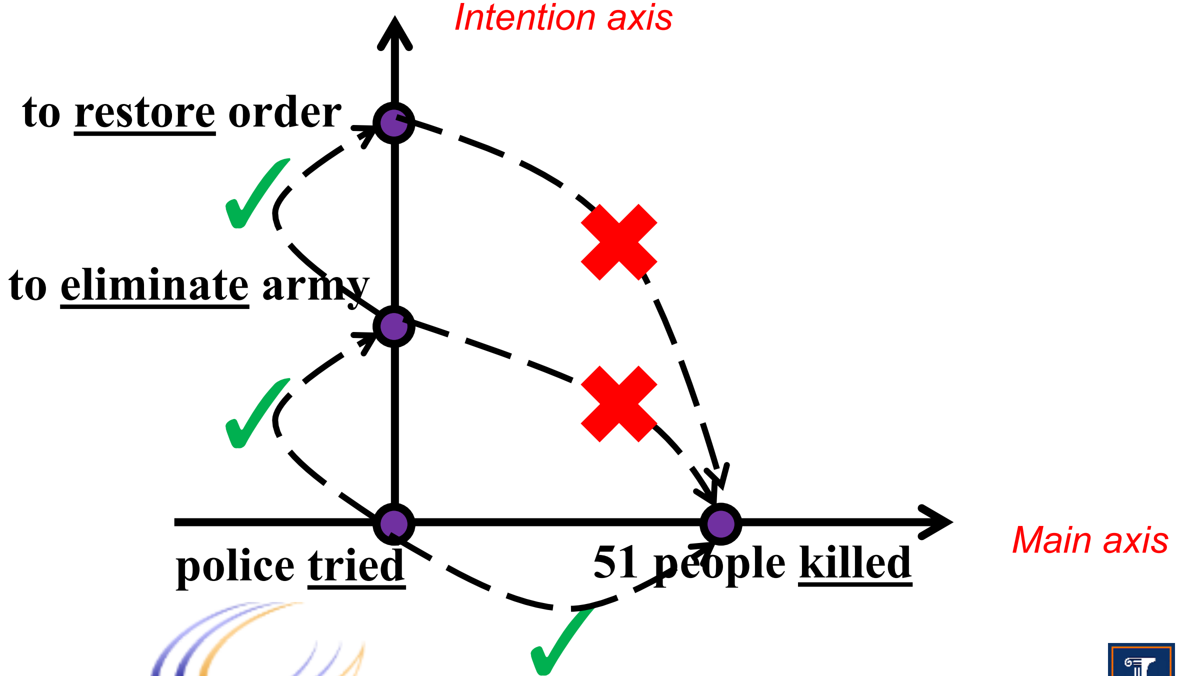
Temporal Structure Modeling: Multi-Axis

- *"Police **tried** to **eliminate** the pro-independence army and **restore** order. At least 51 people were **killed** in clashes between police and citizens in the troubled region."*

- E.g., what's the relation between ***restore*** and ***killed***?
  - restore `BEFORE` killed?
  - restore `INCLUDES` killed?
  - restore `AFTER` killed?

TEMPORAL STRUCTURE MODELING: MULTI-AXIS

- *"Police **tried** to **eliminate** the pro-independence army and **restore** order. At least 51 people were **killed** in clashes between police and citizens in the troubled region."*
- As a balance between "graph" and "chain", we suggest that **multiple temporal axes may exist in natural language**.



MULTI-AXIS MODELING

- We also extend these ideas to more axes

Event Type	Time Axis	%
intention, opinion	orthogonal axis	~20
hypothesis, generic	parallel axis	
Negation	not on any axis	~10
static, recurrent	not considered now	
all others	main axis	~70

Overview: Multi-Axis Annotation Scheme

- Step 0: Given a document in raw text
- Step 1: Annotate all the events
- Step 2: Assign each event an axis (intention, hypothesis, …)
- Step 3: On each axis, perform a "dense annotation" scheme

QUALITY OF OUR NEW DATASET

		Step 2: Axis	Step 3: TempRel
Expert (~400 random relations)		$\kappa = 85\%$	$\kappa = 84\%, F_1 = 90\%$
Crowdsourcing	Accuracy	86%	88%

- Recap: (expert) κ value in TB-Dense is ~60%
- Quality metric for crowdsourcing: Worker Agreement With Aggregate (WAWA) assumes that the aggregated annotations are gold and then compute the accuracy.

Results

- We implemented a baseline system, using conventional features and the perceptron algorithm
- The overall performance on the proposed dataset is <u>much better</u> than those in the literature for TempRel extraction
  - We do NOT mean that the proposed baseline is better than other existing algorithms
  - Rather, the proposed annotation scheme <u>better defines the machine learning task</u>.

Annotation	Training Set	Test Set	Training			Test		
			P	R	F	P	R	F
TB-Dense	Same-axis & Cross-axis	Same-axis	44	67	53	40	60	48
Proposed	Same-axis	Same-axis	**73**	**81**	**77**	**66**	**72**	**69**

[TB-Dense] Taylor Cassidy, Bill McDowell, Nathanel Chambers,and Steven Bethard. "An annotation framework for dense event ordering.". ACL'14.

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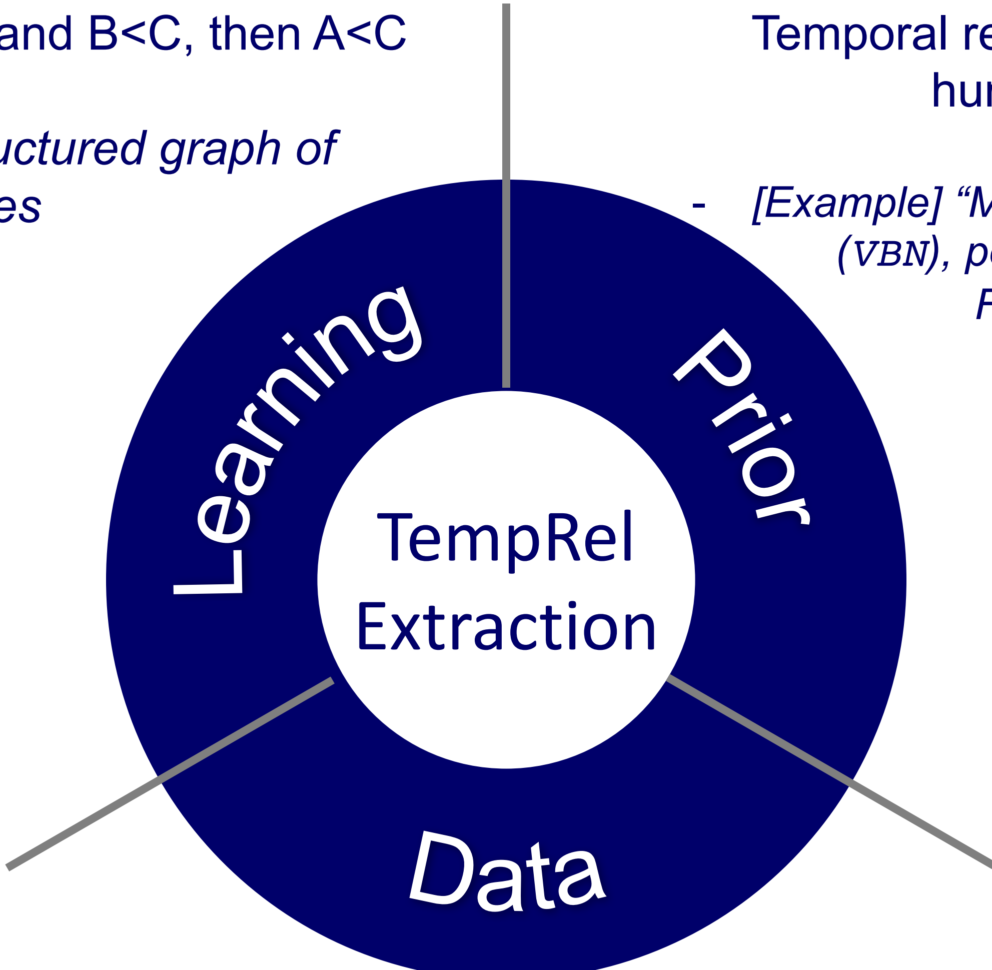


Lack of high-quality data

- *[Intensive labor] Given n events, there're $O(n^2)$ edges*
- *[Difficulty] Not only before/includes, but also involves whether a relation exists*

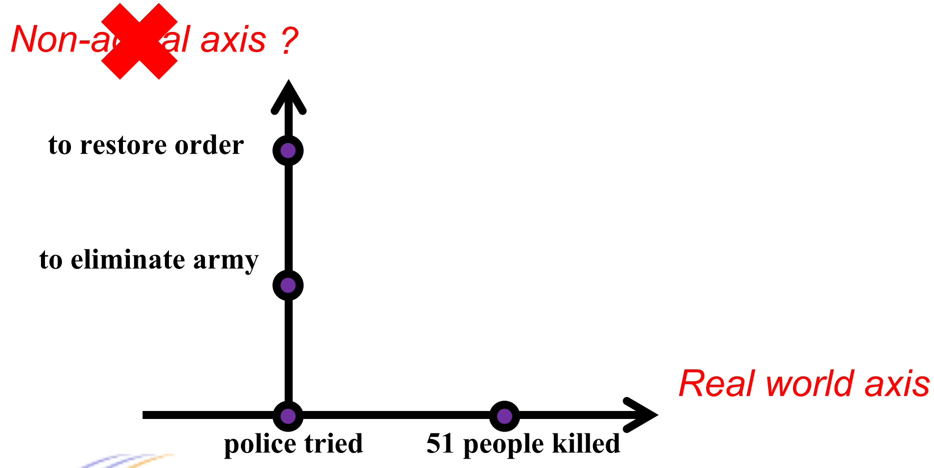
1. Temporal Structure Modeling: Difficulty

- *"Police **tried** to **eliminate** the pro-independence army and **restore** order. At least 51 people were **killed** in clashes between police and citizens in the troubled region."*

- Why is ***restore*** vs ***killed*** confusing?
  - One possible explanation: the text doesn't provide evidence that the ***restore*** event actually happened, while ***killed*** actually happened
  - So, non-actual events don't have temporal relations?
- We don't think so:
  - ***tried*** is obviously before ***restore***: actual vs non-actual
  - ***eliminate*** is obviously before ***restore***: non-actual vs non-actual
  - So relations may exist between non-actual events.

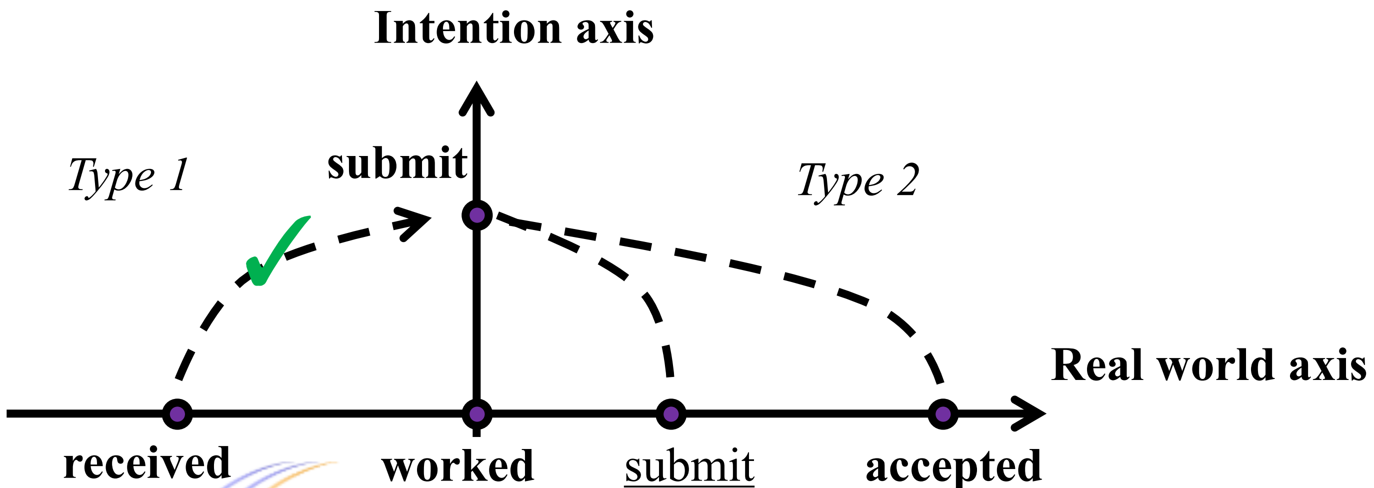
1. Multi-Axis Modeling: NOT Simply Actual Vs Non-Actual

- *"Police **tried** to **eliminate** the pro-independence army and **restore** order. At least 51 people were **killed** in clashes between police and citizens in the troubled region."*
- Is it a "non-actual" event axis?—We think no.
  - First, ***tried, an actual event,*** is on both axes.
  - Second, whether ***restore*** is non-actual is questionable. It's very likely that order was indeed ***restored*** in the end.



CROSS-AXIS RELATIONS

- This work doesn't annotate cross-axis relations for convenience.
- But is it impossible to annotate them?
- *"I **received** positive feedback about my idea. Then I **worked** very hard to **submit** a paper. Finally, the paper was **accepted**."*
  - Type 1: actually covered by our annotation
  - Type 2: ***submit*** has a projection <u>submit</u> onto the real world axis.



- Platform: CrowdFlower
- Quality control: A gold set is annotated by experts beforehand.
  - **Qualification**: Before working on this task, one has to pass with 70% accuracy on sample gold questions.
  - Important: with the older task definition, no crowdsourcers could pass the qualification test.
  - **Survival**: During annotation, gold questions will be given to annotators without notice, and one has to maintain a 70%-accuracy
  - **Majority vote:** At least 5 different annotators are required for every judgement and by default, the majority vote will be the final decision.

3. An Interesting Observation: Ambiguity In End-Points

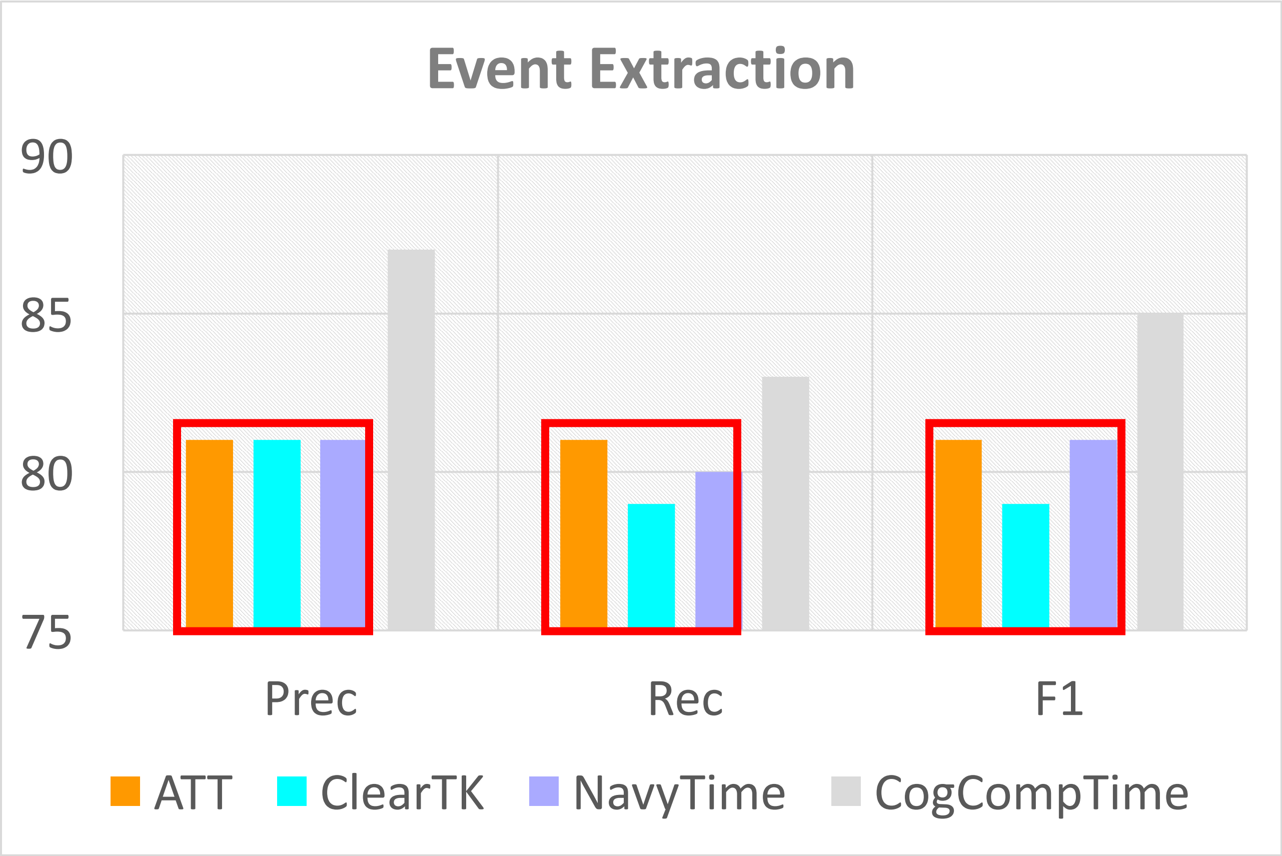
- Given two time intervals: $[t^1_{start}, t^1_{end}], [t^2_{start}, t^2_{end}]$

Metric	Pilot Task 1 t^1_{start} vs t^2_{start}	Pilot Task 2 t^1_{end} vs t^2_{end}	Interpretation
Qualification pass rate	50%	11%	Comparing the end-points is significantly harder than comparing the start-points.
Survival rate	74%	56%	
Accuracy on gold	67%	37%	
Avg. response time	33 sec	52 sec	Task 2 is also significantly slower.

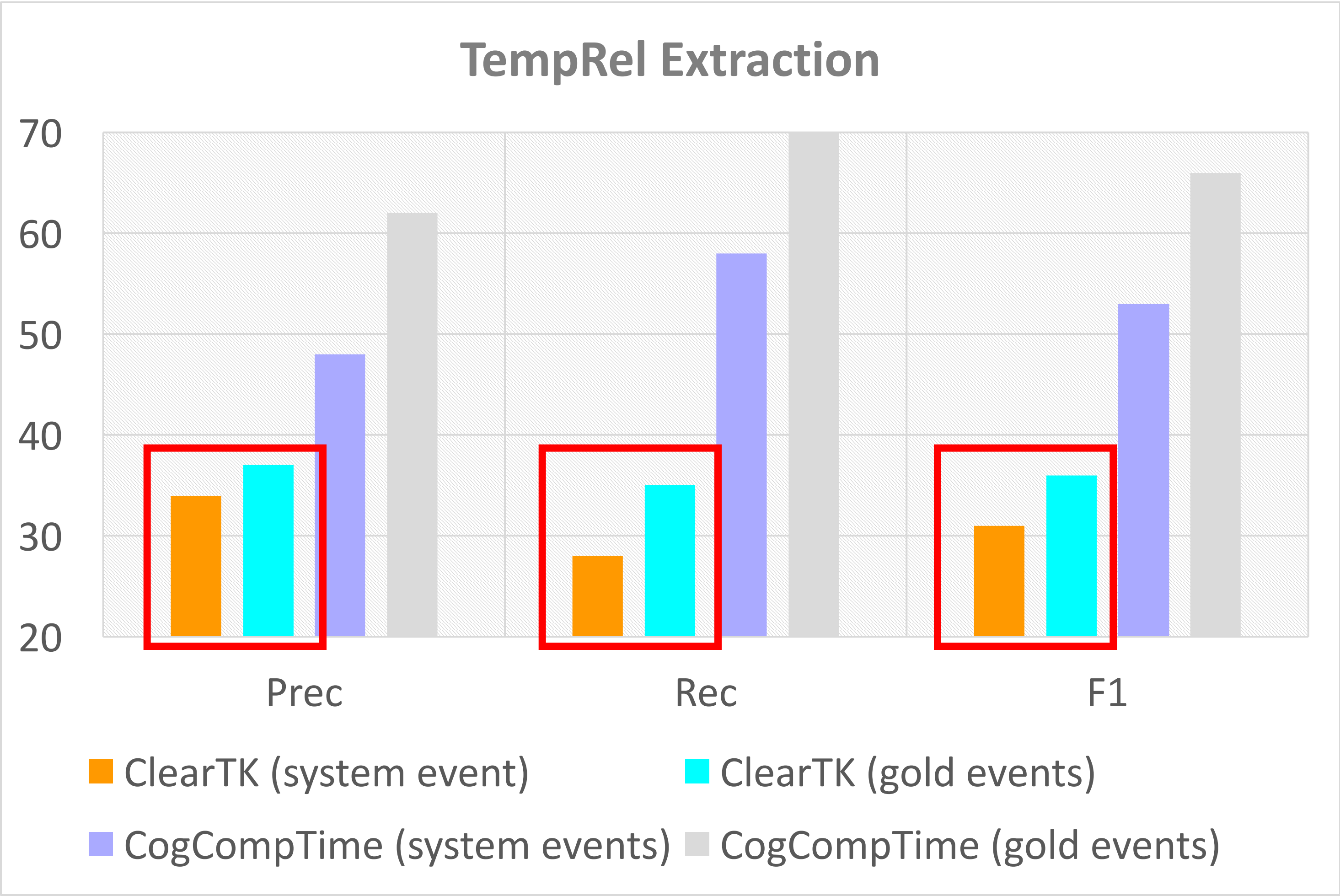
- How durative events are expressed (by authors) and perceived (by readers):
  - Readers usually take longer to perceive durative events than punctual events, e.g., "***restore** order*" vs. "***kill***".
  - Writers usually assume that readers have a prior knowledge of durations (e.g., college takes 4 years and watching an NBA game takes a few hours)
- We only annotate start-points because duration annotation should be a different task and follow special guidelines.

CogCompTime [EMNLP'18]: Put All Pieces Together

- Extraction of (main-axis) events (via a BIO chunker)



CogCompTime [EMNLP'18]: Put All Pieces Together



PROPOSED REMAINING WORK

Thesis Proposal

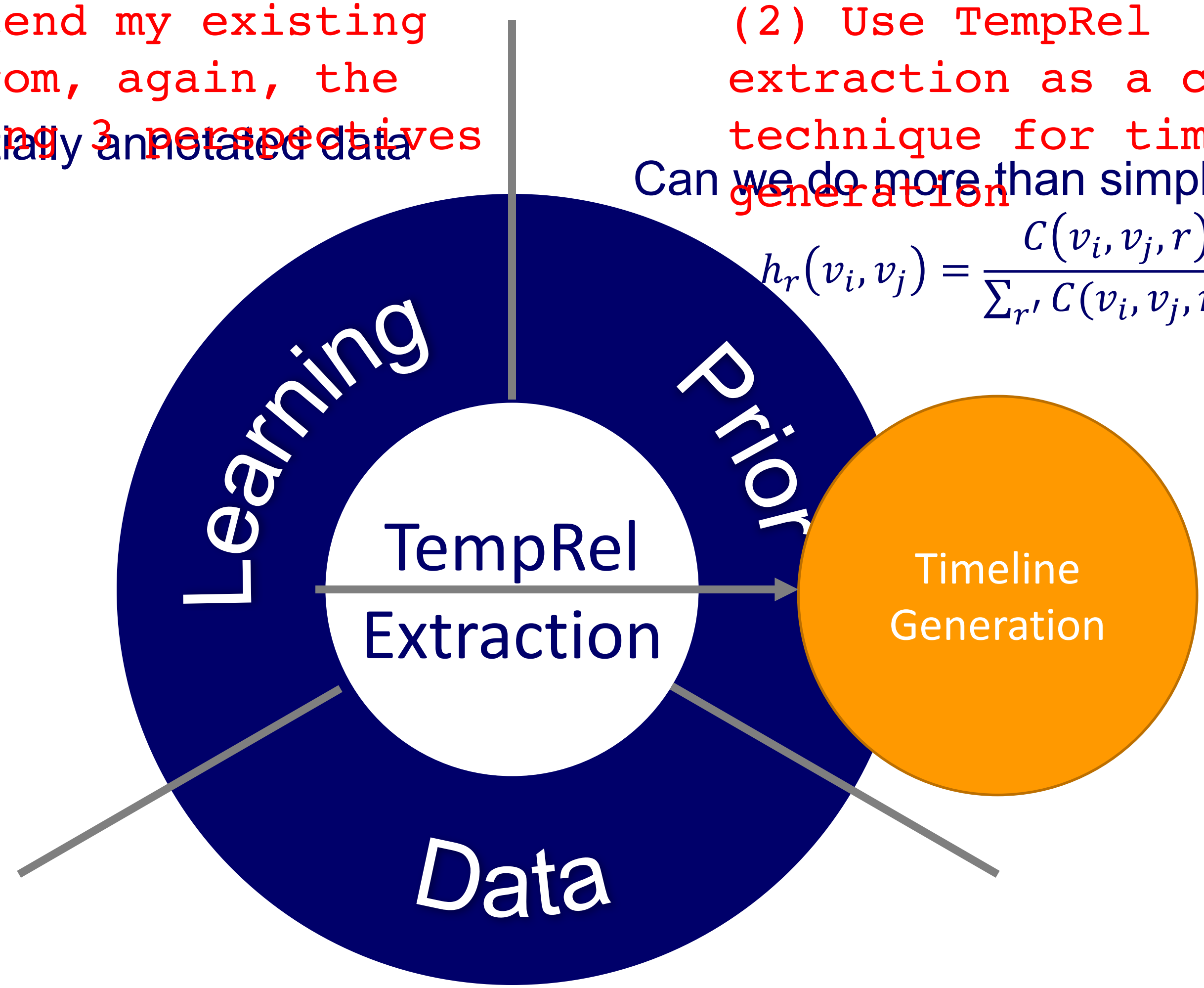
(1) Extend my existing work from, again, the following 3 perspectives

Learning from partially annotated data

(2) Use TempRel extraction as a core technique for timeline generation

Can we do more than simply counting?

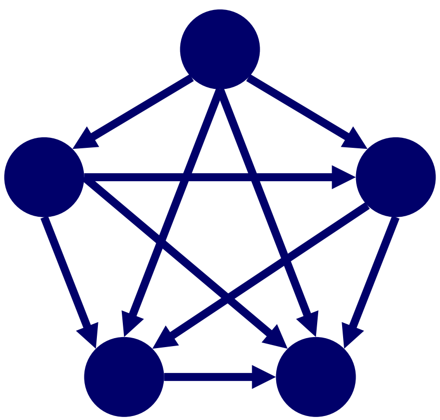
$$h_r(v_i, v_j) = \frac{C(v_i, v_j, r)}{\sum_{r'} C(v_i, v_j, r')}$$



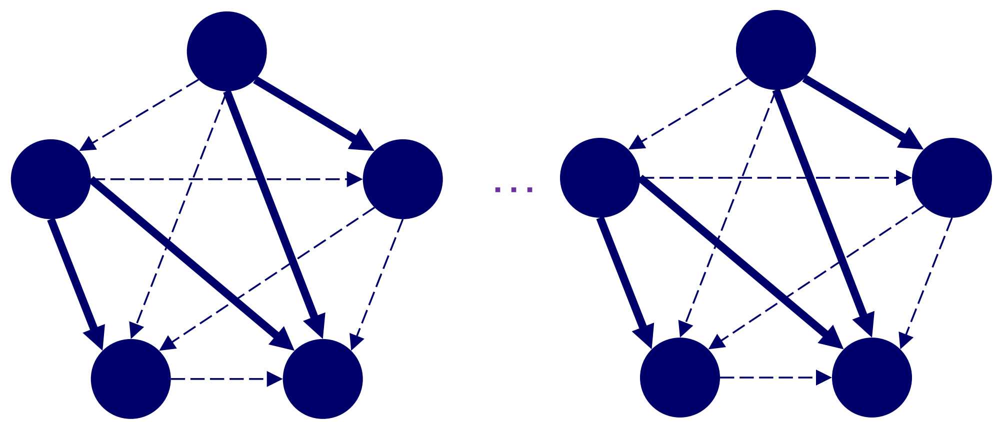
- Further improving crowdsourcing quality
- Extending to nominal events

STRUCTURED LEARNING: MAKE USE OF PARTIAL ANNOTATIONS

- Data annotation for structured tasks is very labor intensive.
- In practice, the number of annotated structures is often limited.
  - One solution: Annotate more structures.
  - Another solution: Make use of partially annotated structures.

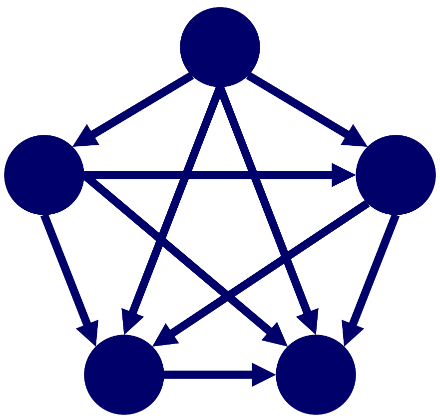


Fully annotated
(F)

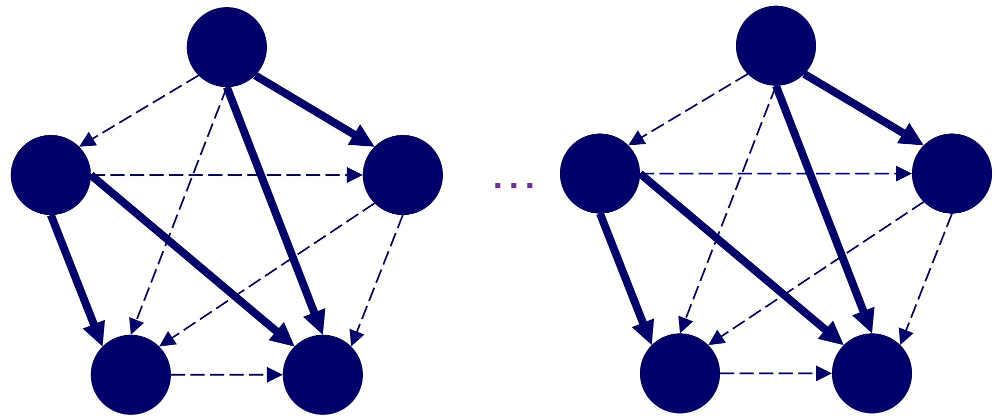


Partially annotated
(P)

Given F and P, How Can We Learn From Them?



Fully annotated
(F)



Partially annotated
(P)

Unannotated
(U)

Let's start from a method, CoDL.

[CoDL] M. Chang, L. Ratinov, and D. Roth. 2007. Guiding semi-supervision with constraint-driven learning. ACL'07.

Constraint-Driven Learning (CoDL): Fully Annotated Data (F) & Unannotated Data (U)

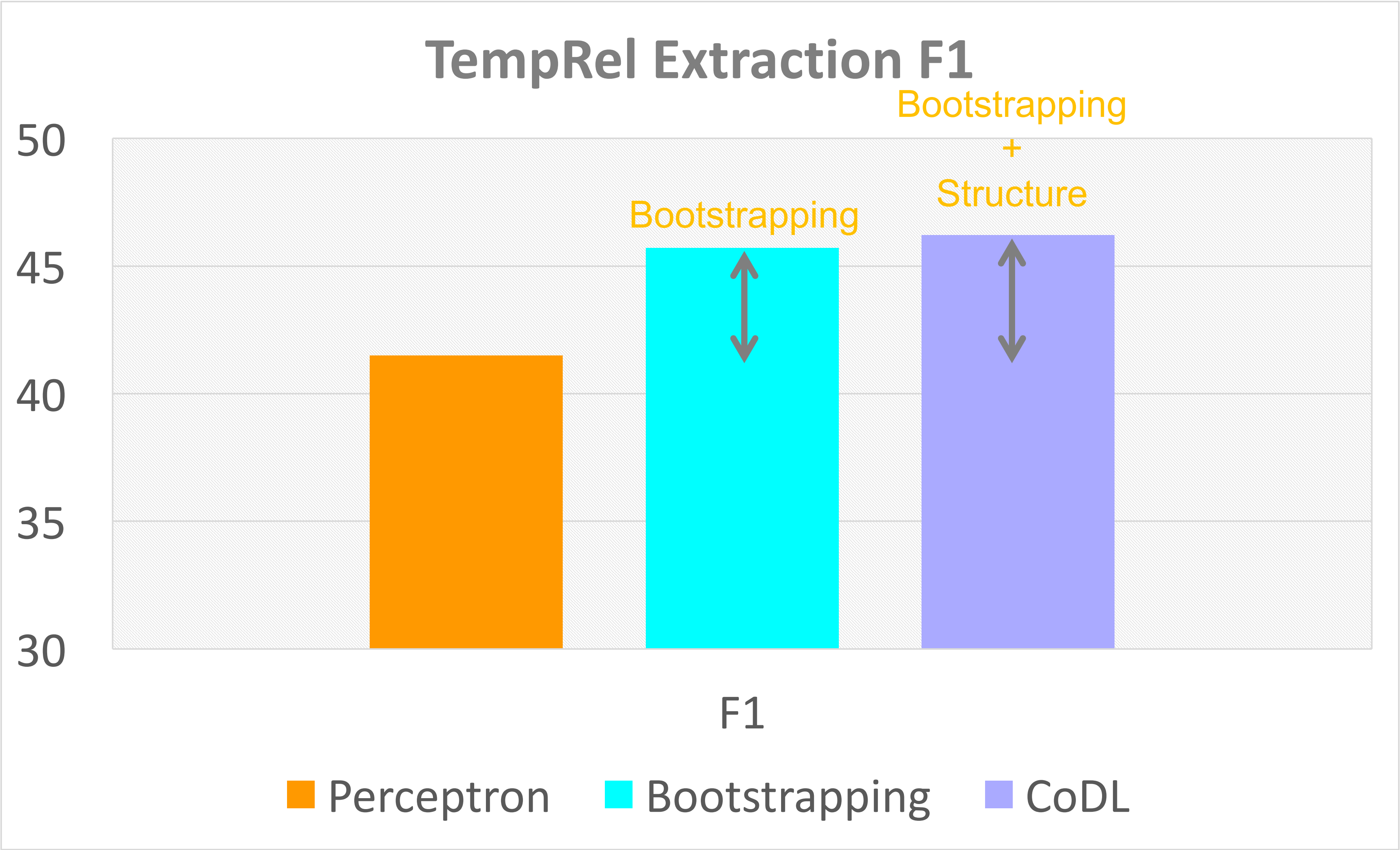
- Initialize the model on F.
- Let the model be S.
- ITERATION

CoDL is a "Constrained bootstrapping".

  - $\widetilde{U} = \emptyset$
  - For each $t \in U$
    - $\hat{y}$ =PREDICT(t) based on S
    - $\widetilde{U} = \widetilde{U} \cup \{(t, \hat{y})\}$
  - S =LEARN($F \cup \widetilde{U}$)

Since t is a structure, when we bootstrap on t, we can enforce the <u>structural constraints</u> of the task at hand; thus the learning is driven by constraints (CoDL).

Back To Our Results On TempRel Extraction



Constraint-Driven Learning (CoDL): Fully Annotated Data & Unannotated Data

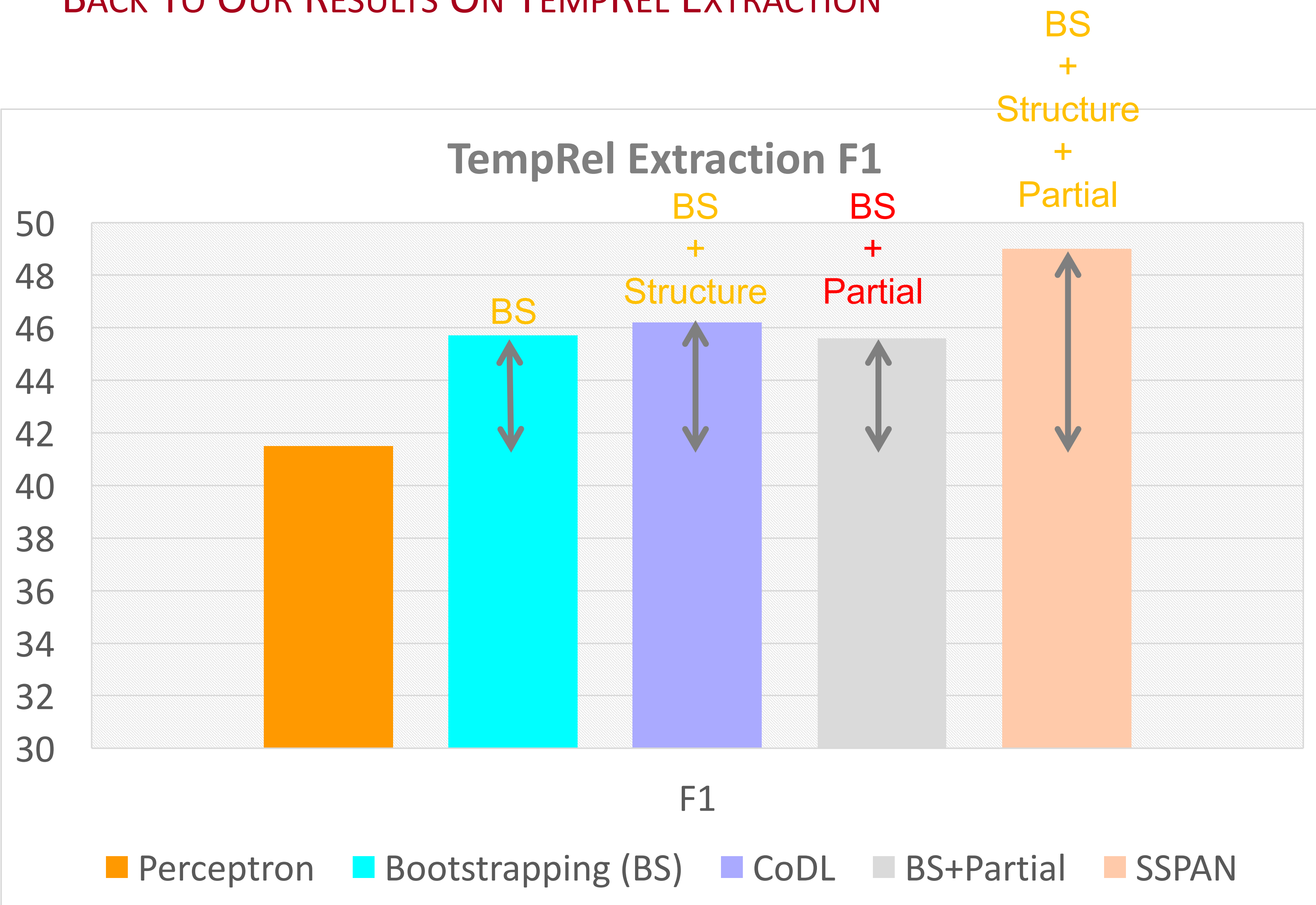
Structured Self-Learning with Partial Annotations (SSPAN) Fully Annotated Data & Partially Annotated Data

- Initialize the model on F. Let the model be S.

- ITERATION
  - $\widetilde{U} = \emptyset$
  - For each $t \in U$
    - $\hat{y}$ =PREDICT(t) based on S
    - $\widetilde{U} = \widetilde{U} \cup \{(t, \hat{y})\}$
  - S =LEARN($F + \widetilde{U}$)

Q: What if t is not completely empty?
A: We can simply add the partial annotations as additional (equality) constraints.

Back To Our Results On TempRel Extraction

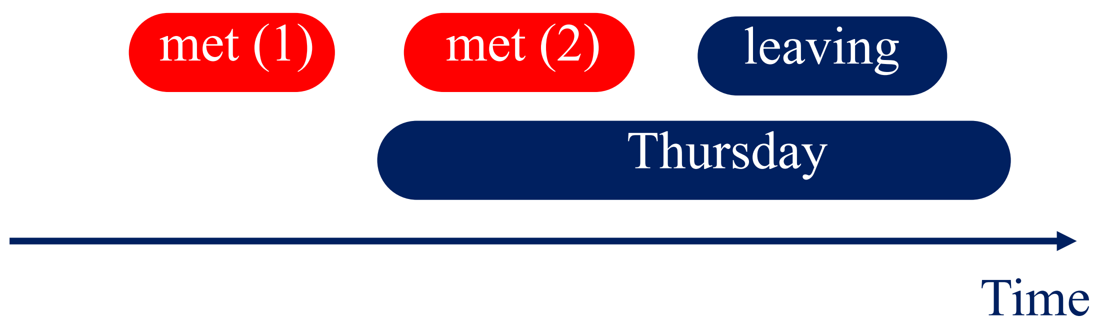


Joint Effect Of Partial & Structure Example 1

Before   Be_Included

I met with him before leaving for Paris on Thursday.

Before ? Be_Included



Joint Effect Of Partial & Structure
Example 2

PREDICATE

The boy gave a frog to the girl.

Arg0 Arg? Arg?

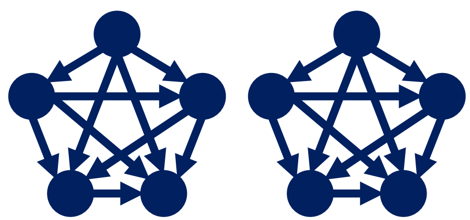
~~Arg0~~ ~~Arg0~~

Cognitive Computation Group

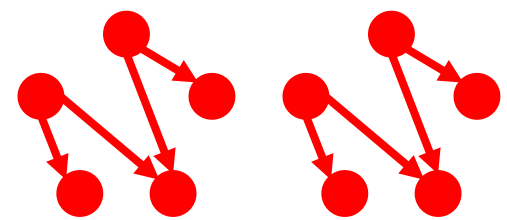
PROPOSAL

- Partial annotations carry information more than themselves in the presence of structural constraints.
- For general structural data collection, which one is better?

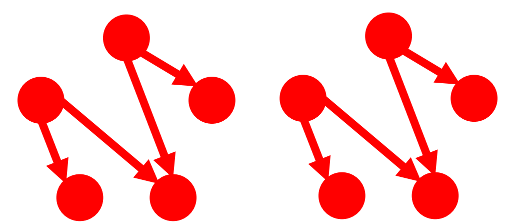
Fully annotated



Partially annotated…

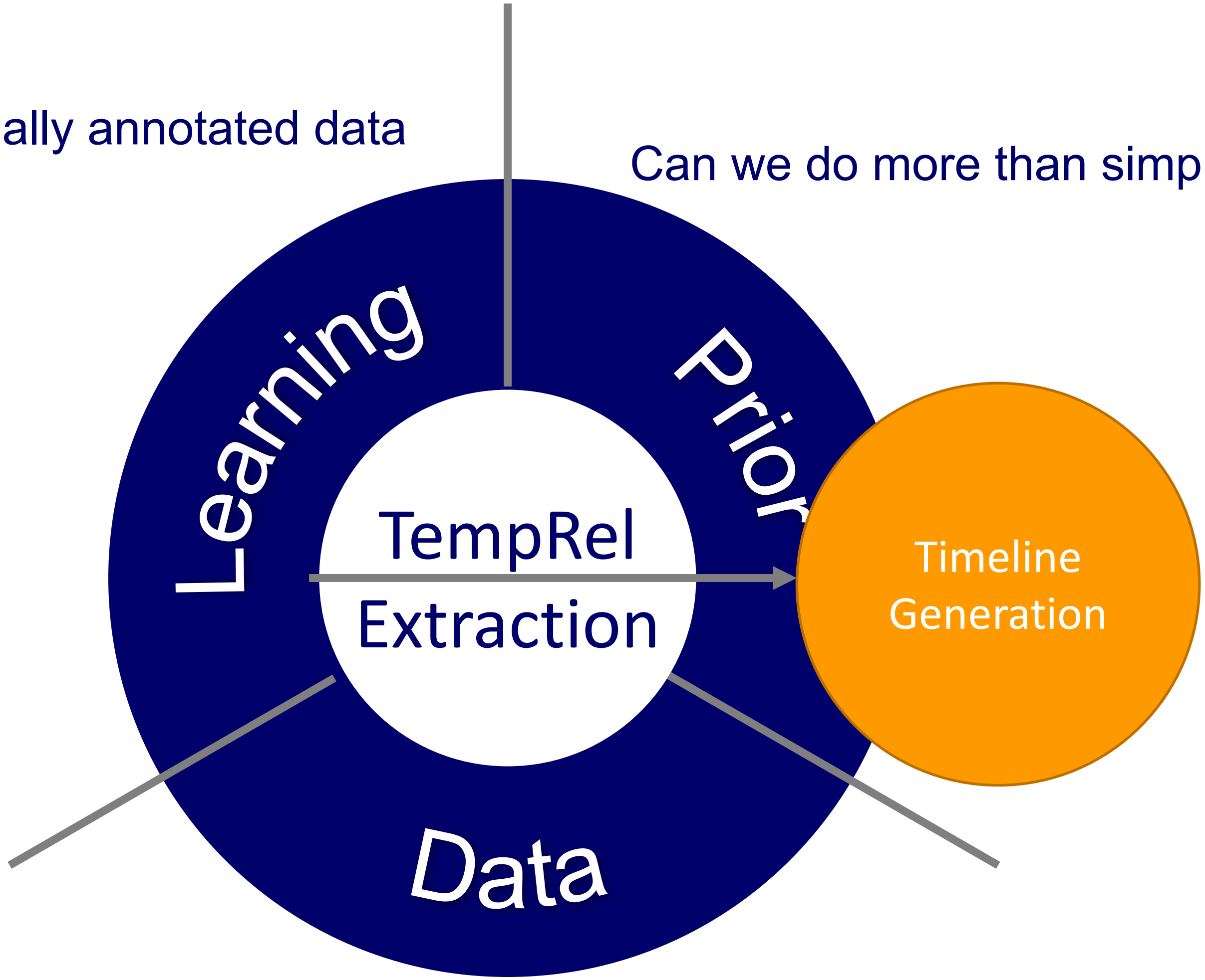


…but more



Learning from partially annotated data

Can we do more than simply counting?



- Further improving crowdsourcing quality
- Extending to nominal events

CogCompTime: Our Online Demo For Temporal

They became friends when they attended the same university 9 years ago. Now they are planning their wedding this June.

The temporal system has identified the following nodes and relations.

They [E0: became] friends when they [E1: attended] the same university [T1: 9 years ago (2009)] . [T2: Now (PRESENT_REF)] they are [E2: planning] their wedding [T3: this June (2018-06)] .

[E0: became] [E1: attended] [T1: 9 years ago (2009)] [T2: Now (PRESENT_REF)] [E2: planning] [T3: this June (2018-06)]

Timeline

```
|
|2009---------E1:attended
|
|------------E0:became
|
|2018-05-15---E2:planning
|
V

Time Axis
```

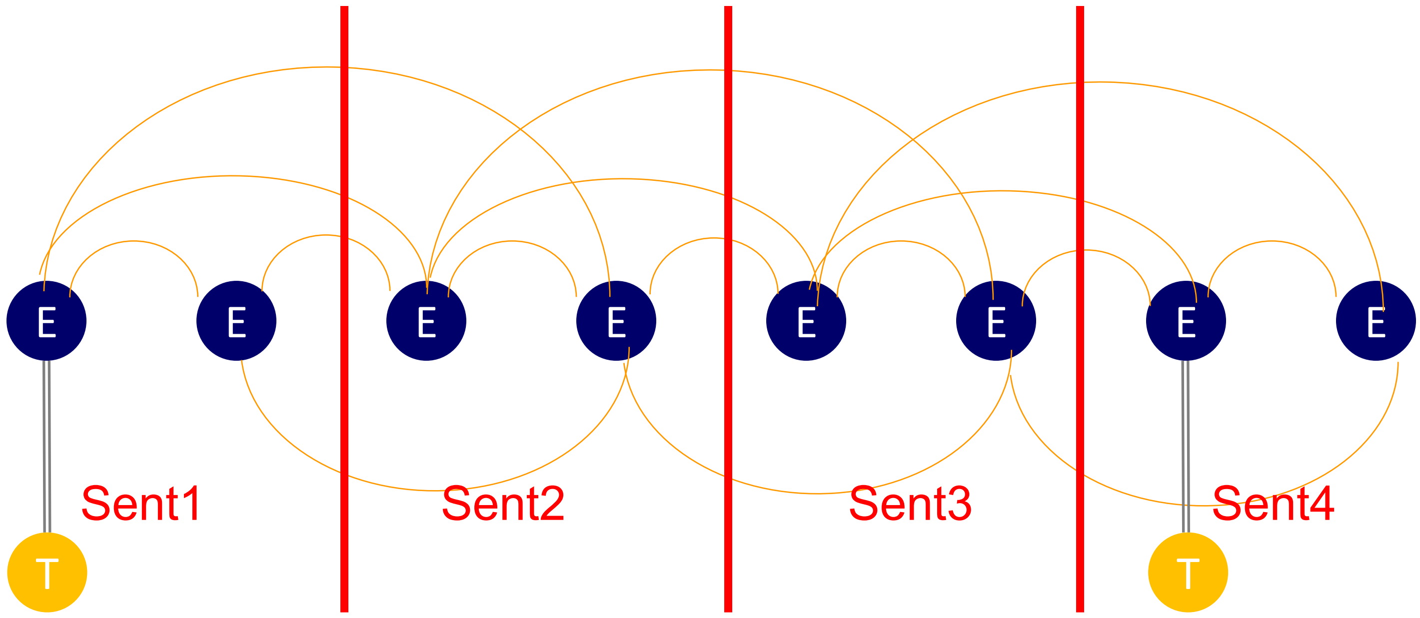
Temporal Graph

This is an early attempt to do timeline generation.
What do I do here?

Previous Formulation of ILP

$$\hat{I} = arg \max_{I} \sum_{i<j} \sum_{r \in \mathcal{R}} f_r(ij) I_r(ij)$$

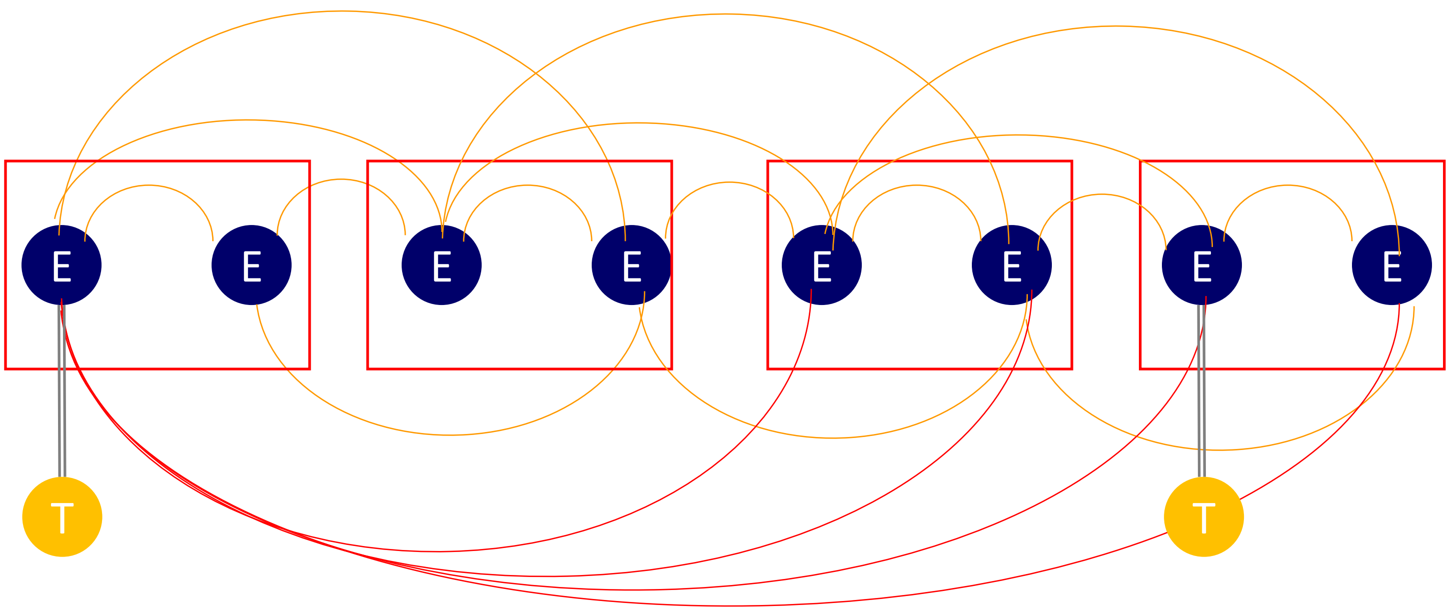
- Events
- Same-sentence pairs
- Cross-sentence pairs
- What if we know the exact time of some events?



This construction of ILP doesn't guarantee the order of time points.

Proposed New Formulation of ILP

- Some time expressions can provide long-distance relations.
  - For example, given two time expressions, “2010” and “2018”, no matter how far they appear in text, there is always a relation that “2010” is before “2018”.



Preliminary Results & Proposal

	P	R	F1
Baseline	62	60	61
+Long-distance	65	73	69

- Long-distance relations are important.
- Proposal: For single document timeline generation, our major missing component is "event coreference", which is another type of long-distance relations.
- For multiple document timeline generation, we are missing
  - Cross-document coreference

SUMMARY

- Understanding time in natural language text
  - …
  - Temporal relation extraction (Focus of my thesis)
  - …
- My past work is from the following 3 perspectives:
  - Structured learning [EMNLP'17, *SEM'18]
  - Prior Knowledge [ACL'18a, NAACL'18]
  - Data annotation [ACL'18b]
- I propose to
  - (1) extend my work from all these perspectives
  - (2) extend my TempRel extraction technique to timeline generation

THANKS FOR YOUR TIME!

And thanks to my little buddy, I was only able to practice my talk once...